

2022 Fall ORF307 1st Midterm Exam Solutions

Problem 1 - Solution

1. We have

$$\begin{aligned}\sum_{i=1}^m w_i (\tilde{a}_i^T x - b_i)^2 &= \sum_{i=1}^m (\sqrt{w_i} (\tilde{a}_i^T x - b_i))^2 \\ &= \sum_{i=1}^m ((D(Ax - b))_i)^2 \\ &= \|D(Ax - b)\|^2\end{aligned}$$

where $D_{ii} = \sqrt{w_i}$. In the second line we use the fact that multiplying a matrix on the left by a diagonal matrix scales the rows of the matrix.

2. To show that the columns of $\tilde{A} = DA$ are linearly independent when the columns of A are, we first observe that column i of $\tilde{A}$ is given by Da_i , where a_i is the i -th column of A . Suppose we have coefficients $\alpha_i, \dots, \alpha_n$ such that

$$\alpha_1 Da_1 + \dots + \alpha_n Da_n = 0.$$

Since the weights are positive, we know that D is invertible. We multiply by D^{-1} on the left of both sides to get

$$D^{-1}(\alpha_1 Da_1 + \dots + \alpha_n Da_n) = D^{-1}0,$$

which simplifies to

$$\alpha_1 a_1 + \dots + \alpha_n a_n = 0.$$

Because the columns of A are linearly independent, only the trivial solution exists, so $\alpha_1 = \dots = \alpha_n = 0$. This means that the columns of $\tilde{A}$ are linearly independent as well.

3. The KKT matrix is given by

$$\begin{bmatrix} 2\tilde{A}^T \tilde{A} & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x^* \\ z^* \end{bmatrix} = \begin{bmatrix} 2\tilde{A}^T \tilde{b} \\ d \end{bmatrix},$$

but we need to rewrite $\tilde{A}$ and $\tilde{b}$ in terms of A , b , and W .

$$\tilde{A}^T \tilde{A} = (DA)^T (DA) = A^T D^T D A = A^T W A$$

since $D^2 = W$ for the diagonal matrix D . We also have

$$\tilde{A}^T \tilde{b} = (DA)^T Db = A^T D^T Db = A^T Wb.$$

Therefore, the final KKT matrix is

$$\begin{bmatrix} 2A^T W A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x^* \\ z^* \end{bmatrix} = \begin{bmatrix} 2A^T W b \\ d \end{bmatrix}.$$

4. The KKT matrix is invertible, as it satisfies both conditions: C has linearly independent rows, and $\tilde{A}$ linearly has independent columns.

Problem 2 - Solution

1. From the linear dynamics, we know that

$$x_2 = Ax_1 + Bu_1$$

For the next step, observe that

$$x_3 = Ax_2 + Bu_2 = A(Ax_1 + Bu_1) + Bu_2 = A^2x_1 + ABu_1 + Bu_2$$

Repeating this iteratively until x_K gives

$$x_K = A^{K-1}x_1 + A^{K-2}Bu_1 + A^{K-3}Bu_2 + \cdots + ABu_{K-2} + Bu_{K-1}$$

Using our definition of C and $\tilde{u}$, the final expression becomes

$$x_K = A^{K-1}x_1 + C\tilde{u}$$

2. Given the definition of $\tilde{u}$, we know

$$\|u_1\|_2^2 + \cdots + \|u_{K-1}\|_2^2 = \|\tilde{u}\|_2^2$$

From part 1, $x_K = A^{K-1}x_1 + C\tilde{u} = 0$, Putting these together gives:

$$\begin{aligned} &\text{minimize} && \|\tilde{u}\|_2^2 \\ &\text{subject to} && C\tilde{u} = -A^{K-1}x_1 \end{aligned}$$

3. For the general constrained least squares problem:

$$\begin{aligned} &\text{minimize} && \|\tilde{A}\tilde{u} - \tilde{b}\|_2^2 \\ &\text{subject to} && \tilde{C}\tilde{u} = \tilde{d} \end{aligned}$$

We know that the KKT linear system is given by

$$\begin{bmatrix} 2\tilde{A}^T\tilde{A} & \tilde{C}^T \\ \tilde{C} & 0 \end{bmatrix} \begin{bmatrix} \tilde{u}^* \\ z^* \end{bmatrix} = \begin{bmatrix} 2\tilde{A}^T\tilde{b} \\ \tilde{d} \end{bmatrix}$$

In our case, $\tilde{A} = I, \tilde{b} = 0, \tilde{C} = C, \tilde{d} = -A^{K-1}x_1$. So, the KKT system is:

$$\begin{bmatrix} 2I & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} \tilde{u}^* \\ z^* \end{bmatrix} = \begin{bmatrix} 0 \\ -A^{K-1}x_1 \end{bmatrix}$$

4. Expanding the KKT system gives the two equations:

$$2\tilde{u}^* + C^T z^* = 0$$

$$C\tilde{u}^* = -A^{K-1}x_1$$

Rearrange the first equation to find $\tilde{u}^* = -\frac{1}{2}C^T z^*$. Plugging this into the second equation gives

$$C(-\frac{1}{2}C^T z^*) = -A^{K-1}x_1 \implies CC^T z^* = 2A^{K-1}x_1$$

Since we assume C is wide with linearly independent rows, we have that CC^T is invertible, so

$$z^* = 2(CC^T)^{-1}A^{K-1}x_1$$

Lastly, plugging this formula for z^* into the first equation and rearranging gives the explicit formula for $\tilde{u}^*$:

$$\tilde{u}^* = -C^T(CC^T)^{-1}A^{K-1}x_1$$

Problem 3 - Solution

1.

$$\begin{aligned} & \text{minimize} && 2t \\ & \text{subject to} && -\mathbf{1}^T x \leq -1 \\ & && -x \leq 0 \\ & && x - t\mathbf{1} \leq 0 \\ & && -x - t\mathbf{1} \leq 0 \end{aligned}$$

2. The feasible region is the space to the upper right of the green lines, inclusive of the lines.

3. The contour lines are given in the plot. For the objective value of one, the contour line consists of a single point.

4. $x^* = (0.5, 0.5)$. It is not possible to achieve $x_1 + x_2 \geq 1$ and $x \geq 0$ if both are less than 0.5. Graphically, we see that each contour plot runs in a box shape from the x_1 -axis to the x_2 -axis because of the infinity norm penalty. From the contours, we can see that it is not possible to achieve a better objective than 1.

5.

$$\begin{aligned} & \text{minimize} && 2(t^+ - t^-) \\ & \text{subject to} && -\mathbf{1}^T x + s_1 = -1 \\ & && x + s_2 - (t^+ - t^-)\mathbf{1} = 0 \\ & && -x + s_3 - (t^+ - t^-)\mathbf{1} = 0 \\ & && x, t^+, t^-, s_1, s_2, s_3 \geq 0 \end{aligned}$$

It is also correct to note that the original t variable is already guaranteed to be non-negative. Therefore there is no need to split it. So the following is also correct.

$$\begin{aligned} & \text{minimize} && 2t \\ & \text{subject to} && -\mathbf{1}^T x + s_1 = -1 \\ & && x + s_2 - t\mathbf{1} = 0 \\ & && -x + s_3 - t\mathbf{1} = 0 \\ & && x, t, s_1, s_2, s_3 \geq 0 \end{aligned}$$